Journal of Computational Science 23 (2017) 192–194

Contents lists available at ScienceDirect

Journal of Computational Science

journal homepage: www.elsevier.com/locate/jocs

Bioinspired algorithms and complex systems☆

ARTICLE INFO

Keywords:
Complex systems
Bioinspired algorithms
Metaheuristics

ABSTRACT

Bioinspired algorithms are search, optimization, and learning techniques whose functioning is based on some metaphor of a biological process. Prominent examples include evolutionary algorithms and swarm intelligence methods. The practical application of these techniques to real-world problems typically involves orchestrating the interplay among different algorithmic components in order to attain synergistic search capabilities. This is a common theme in complex systems, in which the whole is more than the sum of the parts due to the complex interaction patterns among system components, giving rise to emergent properties not anticipated at the base level. Indeed, such systems are prevalent in many contexts, both natural (biological systems, ecosystems, etc.) and artificial (social networks, finance markets, etc.). Analyzing and understanding such systems is not only of the foremost interest, but also constitutes in general a formidable task requiring powerful tools. Metaheuristics in general and bioinspired algorithms in particular can greatly contribute to this end. Furthermore, their intrinsic complex nature makes them prone to be also subject of analysis using a complex-system perspective.

1. Introduction

The term *bioinspired algorithms* [1] refers to metaheuristic methods that draw some inspiration from nature to solve search, optimization or pattern recognition problems; genetic algorithms, for instance, are inspired by Darwin's evolutionary theory [2] whereas ant-colony optimization algorithms are based in the concept of stigmergy [3], introduced to explain self-organization in ant colonies via indirect communication through modification of the environment. Indeed, these are two examples of the two main families of bioinspired metaheuristics, namely evolutionary algorithms [4] (based on the metaphor of evolution, either biological or cultural) and swarm intelligence [5] (based on the collective behavior of decentralized collection of agents, i.e., birds, insects, etc.). Overall, these techniques have set an impressive record of success in solving complex optimization problems.

On the other hand, complex systems are composed of many components, whose structure and interaction lead to emergent properties of the system as a whole, that is, the system exhibits properties and behaviors that are not explicit in its isolated components. This kind of systems are prevalent in natural an artificial contexts, i.e., ecosystems, financial markets, social environments, etc. [6–8]. Complex systems are often at the edge of chaos [9], in the sense that they lie in the middle ground between ordered systems and chaotic ones. This is often the result of the presence of non-linear interactions and feedback loops, often leading to self-organization patterns [10,11].

Bioinspired algorithms and complex systems are connected in many ways. Firstly, it is possible to model bioinspired algorithms using complex systems at a certain level; This is particularly true for hybrid or composite approaches, e.g., memetic algorithms [12], in which different algorithmic components are integrated aiming to obtain an improved emergent search behavior via their synergistic interaction. From a socio-biological standpoint, this ought to be the outcome of competition and collaboration processes among components. In general, the complexity of the interaction structure and the possible presence of uncontrollable changes in the environment underpins the need for using adequate analytical tools. For example, one can follow some well established organization of components/individuals forced by their conditioning and skills or/and by some social tradition. If such structure is observed to be efficient in solving particular group of problems (e.g. surviving) then it might be mapped into the problem area of interest by defining an appropriate complex system of data structures and algorithms.

Recent years have witnessed different attempts to incorporate ideas and/or properties of complex systems into bioinspired algorithms in order to improve their performance or to better understand their behavior. For instance, self-organized criticality [13], self-stabilization [14] or complex topologies [15], just to mention a few. The obverse is also true: complex systems models can also understood, analyzed and studied using bioinspired algorithms, e.g., [16–18]. As a matter of fact, studying complex systems

☆ Carlos Cotta acknowledges support from Spanish Ministry of Economy and Competitiveness and European Regional Development Fund (FEDER) under project EphemeCH (TIN2014-56494-C4-1-P). Robert Schaefer acknowledges support from Polish National Science Centre grant no. DEC-2015/17/B/ST6/01867.

https://doi.org/10.1016/j.jocs.2017.11.010
1877-7503/© 2017 Published by Elsevier B.V.

often requires the use of composite strategies whose components approach the problem (or parts of it) from different perspectives, e.g., [19,20]).

This thematic special issue revolves precisely around the intersection of bioinspired algorithms and complex systems from the two points of view mentioned, namely the use of bioinspired algorithms to tackle and analyzing complex systems, and the design of complex bioinspired algorithms. We have gathered five papers [21–25] targeted to cover algorithmic and implementation aspects of such complex meta-heuristics in both discrete and continuous domains, as well as applications to complex systems. Some contributions to this thematic special issue are extended versions of results communicated at the EvoCOMPLEX track of the EvoApplications conference [26], held in Porto, 30 March – 1 April 2016 as a part of the EvoStar event[1].

2. An overview of this special issue

The papers included in the special issue tackle a variety of topics. Three of them [21,23,25] deal with complex optimization problems and propose suitable bioinspired algorithms for this purpose. Three of them also focus on the algorithmic perspective, be it form an empirical [21,24] or more theoretically-slanted perspective [22]. A more detailed introduction to these papers is provided next.

Gupta and Ahlawat [21] propose a complex bioinspired metaheuristic called Modified Binary Bat Algorithm (MBBA) to select the most essential features characterizing software in the Hierarchic Based Usability Model. The software usability is defined as a quality factor that assesses how easy its interfaces are to use. The model studied is based on twenty three predefined measurable software usability criteria (features). Their MBBA version is able to reduce significantly the number of criteria that have to be measured and processed in order to evaluate the software usability with reasonable accuracy, outperforming previous version of the algorithm.

Karcz-Dulęba and Cichon [22] present the results of an asymptotic analysis of a simple evolutionary algorithm solving global optimization problems in a continuous domain. The behavior of expected trajectories of the associated stochastic dynamic system in the space of its states is studied for small populations composed of two individuals, aiming to explain the phenomena of rapid population's unification and diversification. A stochastic model of expected dynamics also allows reveling bifurcation and chaotic behavior and their dependence on the standard deviation of the probability distribution utilized by the mutation operator. A very interesting result of this paper consists of evaluating the statistics of time to convergence to the steady state of the algorithm, indicating the end of its learning process. The presented study is a valuable attempt to the effective analysis of a population-based complex metaheuristics which is based on their collective, observable and measurable behaviors rather than in local stochastic characteristics.

Poteralski [23] proposes two new complex strategies devoted to the optimal design of the components of a mechanical system. His solutions are based mainly on artificial immune schemes coupled with a novel hyper mutation gradient operation, limited memory Broyden–Fletcher–Goldfarb–Shanno method and Kriging objective approximations. The proposed metaheuristics are firstly tested on standard global optimization benchmarks and then applied to real-world problems in shape design and identification of material properties of mechanical parts, a difficult ill-posed optimization task. After careful tuning, the new strategies exhibit much better efficiency than standard deterministic or stochastic strategies.

Qi, Zhu and Zhang [24] develop new instances of the Artificial Butterfly Optimization (ABO) algorithm, a kind of population-based stochastic global search in which the position of the individual (butterfly) encodes its candidate solution (genotype) and the random mixing operations mimic various flight modes of butterflies to other positions motivated by obtaining food and reproductive goals. The strategy offers an interesting method of balance between exploration and exploitation by dynamically speeding these tasks among two sub-populations of canopy- and sunspot-butterflies respectively. Two new complex ABO strategies – termed ABO1 and ABO2 – were put to test on a benchmark composed of twenty two 30-dimensional continuous functions.

The paper by Turek et al. [25] tackles the problem of urban crossroad management modeled and managed in a micro-scale. Their approach is based on a multi-planning paradigm involved into the Evolutionary Multi Agent complex metaheuristic. The proposed strategy is compared with the results of the A^* algorithm, and is proven to be specially effective when implemented using a functional language (Erlang in this case) and executed in parallel on multi-core hardware.

3. Perspectives

Papers in this special issue provide a good illustration of active research trends in the intersection between complex systems and bioinspired algorithms. The application perspective in undoubtedly crucial. Bioinspired algorithms are the weapon of choice for tackling numerous problems in many domains, and complex systems are no exception. Dealing with such complex problems has been shown to commonly involve the use of a brigade of algorithmic components aimed to tackle the complexity of the problem from different complementary perspectives or dimensions. Complex collective systems such as agent-based evolutionary algorithms or other nature-inspired methods naturally fit into this scenario. Furthermore, when it comes to analyze and design these techniques, we can benefit from the knowledge gained in other areas of complex systems. Chaos theory [27], cellular automata [28] or social computational systems [29] can provide in this sense numerous tools and frameworks for this purpose.

References

[1] S. Olariu, A. Zomaya (Eds.), Handbook of Bioinspired Algorithms and Applications, Chapman & Hall/CRC Computer & Information Science, Boca Ratón FL, 2005.
[2] C. Darwin, On the origin of species by means of natural selection, or the preservation of favoured races in the struggle for life, New York: D. Appleton.
[3] G. Theraulaz, E. Bonabeau, A brief history of stigmergy, Artif. Life 5 (2) (1999) 97–116.
[4] A.E. Eiben, J.E. Smith, Introduction to Evolutionary Computation, Natural Computing Series, Springer-Verlag, Berlin, Heidelberg, 2003.
[5] E. Bonabeau, M. Dorigo, G. Theraulaz, Swarm Intelligence: From Natural to Artificial Systems, Oxford University Press, New York, NJ, 1999.
[6] M. Gell-Mann, Simplicity and complexity in the description of nature, Eng. Sci. 51 (3) (1988) 2–9.
[7] M. Mitchell, Complexity: A Guided Tour, Oxford University Press, Inc., New York, NY, USA, 2009.
[8] M.E.J. Newman, The structure and function of complex networks, SIAM Rev. 45 (2) (2003) 167–256.
[9] C. Langton, Computation at the edge of chaos: phase transitions and emergent computation, Phys. D: Nonlinear Phenom. 42 (1–3) (1990) 12–37.
[10] S.A. Kauffman, The Origins of Order: Self-Organization and Selection in Evolution, Oxford University Press, Inc., New York, 1993.
[11] S. Strogatz, Sync: How Order Emerges From Chaos in the Universe, Nature, and Daily Life, Hachette Books, 2012.
[12] F. Neri, C. Cotta, P. Moscato (Eds.), Handbook of Memetic Algorithms, Vol. 379 of Studies in Computational Intelligence, Springer-Verlag, Berlin, Heidelberg, 2012.

[1] http://www.evostar.org

[13] T. Krink, P. Rickers, R. Thomsen, Applying self-organised criticality to evolutionary algorithms, in: M. Schoenauer, et al. (Eds.), Parallel Problem Solving from Nature PPSN VI, Vol. 1917 of Lecture Notes in Computer Science, Springer, Berlin, Heidelberg, 2000, pp. 375–384.
[14] R. Nogueras, C. Cotta, Studying self-balancing strategies in island-based multimemetic algorithms, J. Comput. Appl. Math. 293 (2016) 180–191.
[15] M. Giacobini, M. Preuss, M. Tomassini, Effects of scale-free and small-world topologies on binary coded self-adaptive CEA, in: J. Gottlieb, G.R. Raidl (Eds.), Evolutionary Computation in Combinatorial Optimization – EvoCOP 2006, Springer, Berlin, Heidelberg, 2006, pp. 86–98.
[16] M. Garbay, M. Rahman, S. Berceli, A multiscale computational framework to understand vascular adaptation, J. Comput. Sci. 8 (2015) 32–47.
[17] B. Liu, S. Koziel, Q. Zhang, A multi-fidelity surrogate-model-assisted evolutionary algorithm for computationally expensive optimization problems, J. Comput. Sci. 12 (2015) 28–37.
[18] M. Vallejo, V. Reiser, D.W. Corne, Genetic algorithm evaluation of green search allocation policies in multilevel complex urban scenarios, J. Comput. Sci. 9 (2015) 57–63.
[19] A.G. Bruzzone, Perspectives of modeling & applied simulation: “modeling, algorithms and simulations: advances and novel researches for problem-solving and decision-making in complex, multi-scale and multi-domain systems”, J. Comput. Sci. 10 (2015) 63–65.
[20] D.E. Keyes, et al., Multiphysics simulations: challenges and opportunities, Int. J. High Perform. Comput. Appl. 27 (1) (2013) 4–83.
[21] D. Gupta, A. Ahlawat, Usability feature selection via MBBAT: a novel approach, J. Comput. Sci. 23 (2017) 195–203.
[22] I. Karcz-Duleba, A. Cichon, Modeling dynamics of small populations in a simple phenotypic evolutionary algorithm a space of population states approach, J. Comput. Sci. 23 (2017) 204–215.
[23] A. Poteralski, Hybrid artificial immune strategy in identification and optimization of mechanical systems, J. Comput. Sci. 23 (2017) 216–225.
[24] X. Qi, Y. Zhu, H. Zhang, A new meta-heuristic butterfly-inspired algorithm, J. Comput. Sci. 23 (2017) 226–239.
[25] W. Turek, L. Siwik, M. Kisiel-Dorohinicki, S. Lakomy, P. Kala, A. Byrski, Real-time metaheuristic-based urban crossroad management with multi-variant planning, J. Comput. Sci. 23 (2017) 240–248.
[26] G. Squillero, et al. (Eds.), Applications of Evolutionary Computation – EvoApplications 2016, Vol. 9597 of Lecture Notes in Computer Science, Springer, Berlin, Heidelberg, 2016.
[27] S. Strogatz, Nonlinear Dynamics and Chaos, Studies in Nonlinearity, Westview Press, Boulder, CO, 2015.
[28] J. Wąs, G.C. Sirakoulis, Cellular automata applications for research and industry, J. Comput. Sci. 11 (2015) 223–225.
[29] N. Agarwal, X. Xu, Social computational systems, J. Comput. Sci. 2 (3) (2011) 189–192.

Carlos Cotta*
Universidad de Málaga, Málaga, Spain

Robert Schaefer**
AGH University of Science and Technology, Kraków, Poland

* Corresponding author.
** Principal corresponding author.
E-mail addresses: ccottap@lcc.uma.es (C. Cotta), schaefer@agh.edu.pl (R. Schaefer).